André Eduardo Meneses do Nascimento

First Break Detection: A Self-Improving Deep Learning Framework

Natal – RN

20th January, 2025

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Monograph presented to the Federal University of Rio Grande do Norte as a partial requirement for obtaining the Bachelor's degree in Computer Engineering

Advisor: Prof. Dr. Tiago Tavares Leite Barros

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Approved on: 20th January 2025

Prof. Dr. Tiago Tavares Leite Barros - Advisor
UFRN

Dr. Ramon Cristian Fernandes Araújo - Co-advisor
UFRN

Prof. Dr. Luiz Affonso Henderson Guedes de Oliveira
UFRN

Prof Dr. Gilberto Corso
UFRN

Natal – RN
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Aos meus pais

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ABSTRACT

This work presents an iterative deep learning framework for first-break detection in seismic data processing that prioritizes dataset-specific adaptability over universal generalization. The methodology combines a U-Net architecture with domain-specific quality control mechanisms to expand the training dataset from limited initial annotations. Experiments conducted on two Canadian mining seismic datasets demonstrate the framework's effectiveness, with the Halfmile dataset showing improvements through iterative training - reducing mean absolute error from 16.16 to 9.16 samples while maintaining high window accuracy (0.9677). Results from the Sudbury dataset reveal the framework's sensibility to data quality and underlying annotation consistency. These findings suggest potential benefits in developing workflows that adapt to specific survey characteristics.

Keywords: deep learning; seismic processing; first breaks; iterative training; quality control.

RESUMO

Este trabalho desenvolve uma estrutura de aprendizado profundo iterativo para detecção de primeiras quebras em processamento sísmico que prioriza adaptabilidade específica ao conjunto de dados ao invés de generalização universal. A metodologia combina uma arquitetura U-Net com mecanismos de controle de qualidade específicos do domínio para expandir o conjunto de treinamento a partir de anotações iniciais limitadas. Os experimentos conduzidos em dois conjuntos de dados sísmicos de mineração canadenses demonstram a eficácia da estrutura, com o conjunto de dados Halfmile apresentando melhorias através do treinamento iterativo - reduzindo o erro absoluto médio de 16.16 para 9.16 amostras enquanto manteve alta precisão de janela (0.9677). Os resultados do conjunto de dados Sudbury revelam a sensibilidade da estrutura à qualidade dos dados e consistência das anotações subjacentes. Estes resultados sugerem benefícios potenciais no desenvolvimento de fluxos de trabalho que se adaptam a características específicas do levantamento.

Palavras-chave: aprendizado profundo; processamento sísmico; primeiras quebras, treinamento iterativo; controle de qualidade.

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LIST OF ABBREVIATIONS AND ACRONYMS

CNN	*Convolutional Neural Network*
SNR	*Signal-to-Noise Ratio*
MAE	*Mean Absolute Error*
GPU	*Graphics Processing Unit*
NPAD	*Núcleo de Processamento de Alto Desempenho*
UFRN	*Universidade Federal do Rio Grande do Norte*
STA/LTA	*Short-Term Average/Long-Term Average*
MER	*Modified Energy Ratio*
QC	*Quality Control*
HDF5	*Hierarchical Data Format version 5*

LIST OF SYMBOLS

y_i	Manual pick for trace i
$\hat{y}_i$	Predicted pick for trace i
σ	Standard deviation
$\mathbb{1}$	Indicator function
N	Total number of traces
A_{exact}	Exact accuracy metric
A_{window}	Window-based accuracy metric
$\lvert \cdot \rvert$	Absolute value
$\mathbf{W}^{(l)}$	Weight matrix at layer l
$\mathbf{h}^{(l)}$	Feature representation at layer l
$\mathbf{b}^{(l)}$	Bias vector at layer l
$\mathcal{L}$	Loss function
$s(t)$	Seismic trace at time t
$w(t)$	Source wavelet at time t
$r(t)$	Earth's reflectivity function at time t
$n(t)$	Noise function at time t
$*$	Convolution operator

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1 INTRODUCTION

The accurate detection of seismic first breaks is a fundamental task in geophysical exploration and data processing. First breaks, which correspond to the initial arrivals of seismic energy at receiver locations, are critical for a variety of preprocessing steps in seismic workflows. These include static corrections, which account for near-surface velocity variations and topographic effects, as well as applications such as velocity modeling and travel time tomography (Marsden, 1993). The accurate identification of these first breaks significantly influences the quality of subsequent seismic imaging and interpretation processes.

Traditional approaches to first-break picking, such as the Short-Term Average/Long-Term Average (STA/LTA) (Coppens, 1985) method and the Modified Energy Ratio (MER) method (Wong, 2016), have demonstrated effectiveness in datasets with high signal-to-noise ratios (SNR). However, these methods face considerable limitations in scenarios with significant noise contamination, weak signals, or complex acquisition geometries. Moreover, the manual refinement of picks in such challenging conditions is an extremely time-consuming process, often requiring days or even months for large-scale seismic datasets. This underscores the limitations of traditional algorithms and highlights the importance of developing automated, robust, and scalable solutions that can significantly reduce human effort and expedite seismic data processing workflows.

Recent advancements in machine learning, particularly in the domain of deep learning (see Goodfellow et al. (2016)), have provided new tools for addressing these challenges. Convolutional Neural Networks (CNNs) (O'Shea and Nash, 2015), with their ability to learn spatial hierarchies and capture intricate patterns, have become a cornerstone for various tasks in seismic data analysis. Among these, the U-Net architecture (Ronneberger et al., 2015) has emerged as a particularly effective solution due to its encoder-decoder structure with skip connections. Originally designed for biomedical image segmentation, the U-Net's ability to integrate fine-grained spatial details with broader contextual information has proven highly valuable in the seismic domain. Nevertheless, while these methods have introduced substantial improvements, issues such as generalization across datasets, noise robustness, and the need for large volumes of labeled data remain key challenges.

To address these challenges, this work presents an iterative framework for first-break detection based on a U-Net architecture. This framework is designed to operate effectively with limited annotated data, leveraging iterative self-training and domain-specific quality control mechanisms to expand the training dataset progressively. By incorporating preprocessing strategies tailored to seismic data characteristics and systematic quality

control of model predictions, this framework aims to improve the accuracy and robustness of automated first-break picking.

1.1 Outline of the Work

This monograph is organized into five chapters, each addressing a specific aspect of the research.

- **Chapter 1** provides an introduction to the problem of first-break detection, outlining its importance in seismic data processing and the limitations of traditional and contemporary methods. It also presents the motivation for this study and the proposed solution.
- **Chapter 2** reviews the theoretical foundations and existing literature relevant to first-break picking. This includes a discussion on machine learning and deep learning techniques, an overview of seismic data acquisition and processing, and a critical analysis of traditional and modern approaches to first-break picking.
- **Chapter 3** describes the methodology adopted in this work. It details the U-Net architecture employed, the preprocessing steps tailored for seismic data, and the iterative training workflow. Furthermore, the chapter outlines the experimental setup and the evaluation metrics used to assess the performance of the proposed framework.
- **Chapter 4** presents the results obtained through the proposed framework, along with a discussion of their implications. The chapter evaluates the framework's effectiveness in improving first-break detection and highlights its strengths and limitations.
- **Chapter 5** concludes the monograph by summarizing the key findings, discussing the broader implications of this research, and suggesting directions for future work.

Through this structure, this work aims to provide a comprehensive exploration of the proposed framework for first-break detection, situating it within the broader context of seismic data processing and machine learning.

2 LITERATURE REVIEW AND THEORETICAL FOUNDATION

This chapter presents the theoretical foundation and literature review necessary to understand and contextualize our work. We begin by introducing neural networks and their evolution toward specialized architectures for spatial data analysis. We then provide an overview of seismic data acquisition and processing, with particular focus on the challenges that motivate our research. Finally, we review the development of first break picking methods, from traditional approaches to current deep learning solutions, identifying limitations and opportunities that inform our methodology.

2.1 Machine Learning and Deep Neural Networks

Computer science emerged from the systematic study of computation and algorithmic abstraction—the practice of decomposing problems into discrete, well-defined procedures. This foundation rests on the premise that problems can be solved through deterministic sequences of operations. However, a significant class of computational challenges, including pattern recognition and natural language understanding, resist traditional algorithmic decomposition due to their inherent complexity and variability. These challenges motivated the development of statistical learning approaches.

The central problem in supervised learning is fundamentally one of function approximation: given a set of input-output pairs $(x_1, y_1), \ldots, (x_n, y_n)$, the objective is to construct a mapping $f : \mathcal{X} \to \mathcal{Y}$ that both fits the observed data and generalizes effectively to new instances. This framing transforms the challenge from explicit programming to learning from examples, where the goal is to discover underlying patterns rather than specify them directly.

The theoretical and practical pursuit of this question has produced a rich framework of methods,from support vector machines to deep neural networks, that now serve as fundamental tools across disciplines. These approaches have proven particularly powerful for problems characterized by high dimensionality, noise, and complex non-linear relationships, enabling advances in fields ranging from computer vision to natural language processing.

Neural networks emerged as a particularly powerful solution framework for these pattern recognition challenges. The architecture consists of hierarchical layers of computational units connected through learnable parameters $\mathbf{W}^{(l)}$ at each layer l. Each layer implements a transformation of the form $\mathbf{h}^{(l+1)} = \sigma(\mathbf{W}^{(l)}\mathbf{h}^{(l)} + \mathbf{b}^{(l)})$, where σ is a nonlin-

ear activation function. This sequential composition of linear and nonlinear operations progressively projects the input data into increasingly abstract feature spaces, enabling the network to discover complex hierarchical patterns. The parameters are optimized through gradient descent on a loss function $\mathcal{L}$, defined as the discrepancy between network predictions and target outputs: $\min_{\mathbf{W},\mathbf{b}} \mathcal{L}(\hat{y}, y)$ (Goodfellow et al., 2016).

The expressive power of deep architectures through hierarchical representations introduces fundamental challenges in model design. While increased model complexity enables the capture of more intricate patterns, it imposes greater computational demands and heightens the risk of overfitting—where the model learns spurious correlations specific to the training data rather than generalizable relationships. This tension between model complexity and generalization capability, formally characterized as the bias-variance trade-off, motivated the development of specialized architectures optimized for specific data modalities.

A significant architectural innovation emerged in the form of Convolutional Neural Networks (CNNs), which exploit spatial structure through local connectivity patterns and parameter sharing. Formally, each convolutional layer implements the operation $h_{i,j}^{(l+1)} = \sigma(\sum_{m,n} W_{m,n}^{(l)} h_{i+m,j+n}^{(l)})$, where the weights $W_{m,n}^{(l)}$ are shared across spatial locations. This inductive bias enables efficient learning of hierarchical features while preserving spatial relationships (Prince, 2023). In seismic data analysis, CNNs have demonstrated superior performance across multiple tasks: from denoising and interpolation (Yu and Ma, 2021) to the identification of geological structures such as faults, horizons, and salt bodies (Mousavi et al., 2023).

The U-Net architecture (Ronneberger et al., 2015) addresses a critical limitation in traditional CNNs: the trade-off between spatial resolution and semantic context. Its symmetric encoder-decoder structure, augmented with skip connections $s^{(l)} : h_{enc}^{(l)} \to h_{dec}^{(l)}$, enables the integration of multi-scale features. This design preserves fine-grained spatial details while incorporating broader contextual information—a capability particularly crucial for semantic segmentation tasks. Originally developed for biomedical image segmentation, the architecture's effectiveness in maintaining both local precision and global context has led to its widespread adoption across domains, including seismic interpretation.

2.2 Reflection Seismology

Reflection seismology forms the foundation of subsurface imaging. The method relies on analyzing elastic or acoustic waves that propagate through the Earth: when these waves encounter interfaces between materials with contrasting acoustic impedances, they partition into transmitted and reflected components. The reflected waves are recorded by surface sensors as time-series data, with each individual recording termed a seismic

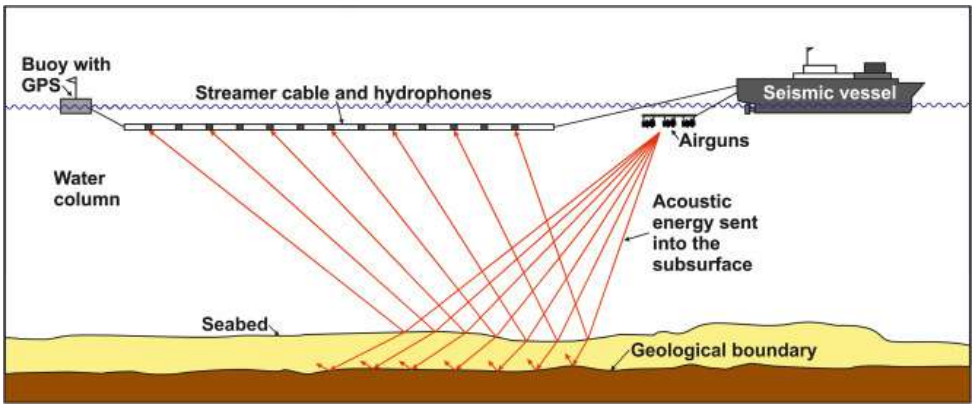

Figure 1 – Marittime Acquisition Illustration - Source: Cox et al. (2020)

trace. Mathematically, a seismic trace $s(t)$ can be expressed as the convolution of the source wavelet $w(t)$ with the Earth's reflectivity function $r(t)$, plus noise $n(t)$: $s(t) = w(t) * r(t) + n(t)$ (Biloti, 2024)

Data acquisition geometries are optimized according to survey objectives and environmental constraints. Land surveys typically employ linear receiver arrays with multiple source points, while marine surveys utilize vessel-towed hydrophone streamers. The spatial sampling interval between receivers, determines the horizontal resolution of the wavefield reconstruction. Similarly, the temporal sampling rate, typically in milliseconds, must be sufficient to capture the highest frequencies of interest.

The recorded data is organized into gathers based on common acquisition parameters, such as source location or reflection point. However, raw seismic records are contaminated by various noise sources: near-surface effects, ambient noise, and multiple reflections all obscure the primary reflections of interest (Biloti, 2024; Yilmaz, 2001). Modern surveys generate massive datasets as acquisitions routinely produce hundreds of thousands of traces. Thus, the scale and complexity of seismic data, combined with the presence of various noise types, demands for extensive pre-processing and processing steps to obtain accurate subsurface images.

2.3 First Break Picking

First breaks in seismic data represent the initial arrival times of seismic energy at receiver locations. Their accurate identification is crucial for seismic data pre-processing, particularly in land acquisitions where they enable static corrections—time shifts applied to each trace to compensate for near-surface velocity variations and topographic effects (Marsden, 1993). First-break times also support various applications including the modeling of water velocity variations (Mardan et al., 2022) and the refinement of velocity models for top salt structures in deep-water reservoirs through travel time tomography (Alves

et al., 2024).

2.3.1 Traditional Methods

The increasing scale of seismic surveys has motivated the development of automated first-break picking methods. Early approaches focused on detecting abrupt changes in signal characteristics. The Short-Term Average/Long-Term Average (STA/LTA) method (Coppens, 1985) compares energy ratios between short and long time windows to identify sudden amplitude variations. While these methods perform adequately on data with high signal-to-noise ratio (SNR), their reliability deteriorates in noisy conditions due to their susceptibility to spurious amplitude spikes.

Further developments sought to enhance robustness to noise. The Modified Energy Ratio (MER) method (Wong, 2016) incorporates preprocessing techniques such as multi-trace summation to improve SNR before applying the picking algorithm. While MER shows improved performance under moderate noise conditions, it remains susceptible to errors when processing weak signals or data with severe noise contamination. An alternative approach based on fractal analysis (Boschetti et al., 1996) identifies first breaks by detecting variations in fractal dimension along seismic traces, attempting to capture the transition from noise to signal. However, these methods, despite their sophistication in emulating human pattern recognition, impose significant computational costs while still exhibiting limitations in low SNR scenarios

Despite these advancements, a significant limitation shared by many of these algorithms is their reliance on processing traces individually, which overlooks the spatial coherence between adjacent traces and exacerbates inaccuracies under challenging conditions. Consequently, automated picking methods often struggle to achieve satisfactory results when dealing with low-quality or noisy data, requiring substantial manual intervention to correct errors and refine picks. As a result, a considerable amount of human effort is still necessary to ensure reliable first-break picking in complex datasets.

2.3.2 Deep Learning approaches

Hence, first break picking has emerged as a domain where the sophisticated pattern recognition capabilities of modern machine learning techniques prove particularly advantageous, as demonstrated by numerous studies evaluating their effectiveness. The problem is commonly formulated as a semantic segmentation task, where each seismogram is processed as an image whose pixels are classified into predefined classes.

St-Charles et al. (2023) established a benchmark for first break picking on publicly available seismic data using U-Nets. Their work demonstrated the generalizability of this approach by conducting training, validation, and testing across different seismic sites. The

study included hyperparameter optimization, evaluation of various encoding/decoding paths, and identification of optimal combinations for each fold. Similarly, Zwartjes and Yoo (2022) evaluated different U-Net architectural parameters, including network depth, loss functions, and input sizes.

Loginov et al. (2022) introduced a 1D convolutional network workflow that achieves comparable results to U-Net while requiring fewer computational resources and only thousands of traces for training. However, the applicability of their model across different seismic settings remains uncertain, as 1D CNNs cannot capture the spatial context that is crucial in low signal-to-noise ratio (SNR) scenarios, where individual traces may become indistinguishable from noise without considering their neighboring traces.

Both Li et al. (2024) and Yuan et al. (2020) propose more sophisticated workflows. Li et al. (2024) employs a parallel structure combining a U-Net and a denoising convolutional network to implement a fine-tuning approach that incorporates knowledge of noisy data. Yuan et al. (2020) introduces a dual-model architecture comprising a U-Net for semantic segmentation and a recurrent neural network for selecting first picks among the inferred classes. The authors develop a straightforward transfer learning scheme where weights from a previously trained network are applied when training on different field data geometries, using only a few newly annotated shots. Their results demonstrate that this approach yields improved evaluation metrics compared to naive weight initialization.

Similarly, Mardan et al. (2024) explored transfer learning by fine-tuning pre-trained ImageNet networks with minimal new annotated examples. However, while their results suggested improved performance through pre-training, the validation was limited by both a small test set and ambiguous evaluation metrics.

These approaches aim to leverage knowledge gained from networks trained on initial datasets through weight fine-tuning, thus minimizing the need for new manual annotations. This strategy for generalization differs fundamentally from St-Charles et al. (2023)'s approach of learning generalizable features from one dataset for direct application to others. However, both methodologies require substantial previously annotated gathers for initial training. For such approaches to be efficiently employed, their generalization capacity must be robust enough to minimize further intensive data preparation needs. Proper validation of these methods would require testing on large volumes of fully annotated gathers—a challenge given the typical access restrictions on seismic data. Nevertheless, St-Charles et al. (2023) advances this effort by releasing labeled data for four distinct field datasets.

Beyond architectural choices and training strategies, the reliability of automated picking methods heavily depends on their ability to validate their own predictions. Several approaches have incorporated quality control mechanisms to evaluate and refine picks.

Loginov et al. (2022) implements a straightforward statistical filtering method, grouping picks by offset bins and discarding those outside a three σ window, using the proportion of retained picks as an indicator of model robustness. Taking a more comprehensive approach, Pu et al. (2024) developed a three-step validation process: statistical filtering, trace coherence assessment using Dynamic Time Warping (Müller, 2007), and cross-validation of predictions across different trace representations. These quality control methods provide valuable performance insights beyond traditional metrics, suggesting that robust automated workflows should incorporate systematic validation mechanisms. Such validation strategies, along with fundamental design choices like classification schemes and data preprocessing approaches, form the foundation of an effective first break picking system, aspects we will examine in detail in our methodology

Aligning with Loginov et al. (2022), we argue that first break picking models should not prioritize high generalizability. Although CNNs exhibit translation and rotation invariance, they lack scale invariance. The model filters learn features specific to their training seismograms' resolution and therefore struggle to generalize across geometries with varying trace counts per gather and sampling rates. While theoretically possible to learn scale-invariant features by training on diverse datasets with different resolutions, this would require more complex architectures capable of learning a broader feature set, substantially increasing computational demands. Moreover, even such enhanced models could fail when encountering significantly different resolutions. This practical limitation is supported by Zwartjes and Yoo (2022), who found that network performance improves as input sizing approaches original dimensions. Furthermore, seismic images lack the rich, distinctive details that enable convolutional models to capture universal features, as they do with natural images

Therefore, we propose that a reliable automated workflow for first break picking should emphasize a precise framework capable of generalizing picks from limited examples, while incorporating robust validation mechanisms to ensure reliability. Such an approach would aim to reduce manual labeling requirements through effective use of limited training data and systematic quality control of predictions.

The developments reviewed in this chapter highlight the evolution of first break picking methods from simple signal processing approaches to sophisticated deep learning solutions. While each advancement has brought valuable insights, significant challenges remain in balancing computational capability with practical constraints. Traditional methods demonstrated the importance of robust signal detection but struggled with noise and trace independence. Modern deep learning approaches showcase the power of spatial context and pattern recognition but face challenges in generalization and data requirements. These observations have guided the development of our methodology, presented in the following chapter, which aims to address these challenges through a focused approach to

training and validation.

3 METHODOLOGY

The methodology presents an iterative framework for training a U-Net model with limited annotated seismic data. The framework integrates preprocessing, model architecture, and a self-training workflow to address the challenge of limited labeled data in seismic interpretation. This approach combines data normalization techniques from computer vision with domain-specific quality controls for seismic data, enabling progressive expansion of the training dataset through controlled self-labeling.

3.1 Framework Overview

The proposed iterative training framework consists of several interconnected stages, as illustrated in Figure 2. The workflow begins with dataset and model initialization, followed by the selection of initial training gathers. These gathers are split into training and validation sets to provide the foundation for the first training cycle.

The core of the framework is an iterative loop that progressively expands the labeled dataset. Each iteration consists of three main phases: model training, inference on new gathers, and quality control. After training, the model selects new gathers for inference based on their spatial distribution and signal quality. These predictions then undergo a two-stage quality control process.

The quality control process begins with a confidence filtering step, where predictions are evaluated based on their softmax probabilities. Subsequently, a tau-offset analysis ensures spatial coherence of the picks within each gather. Gathers that pass both quality control stages have their picks incorporated into the training dataset, while failed gathers are discarded from the current iteration.

After each iteration, the framework checks dataset coverage to determine if the desired proportion of total gathers (50% in our experiments) has been used in training, validation, or inference. If this threshold has not been reached, the process continues with updated training and validation sets incorporating the newly validated picks.

The final evaluation phase assesses model performance specifically on gathers that were not part of the initial labeled set. This provides an assessment of how well the model learned to replicate human picking behavior using both the original manual picks and the iteratively added predictions.

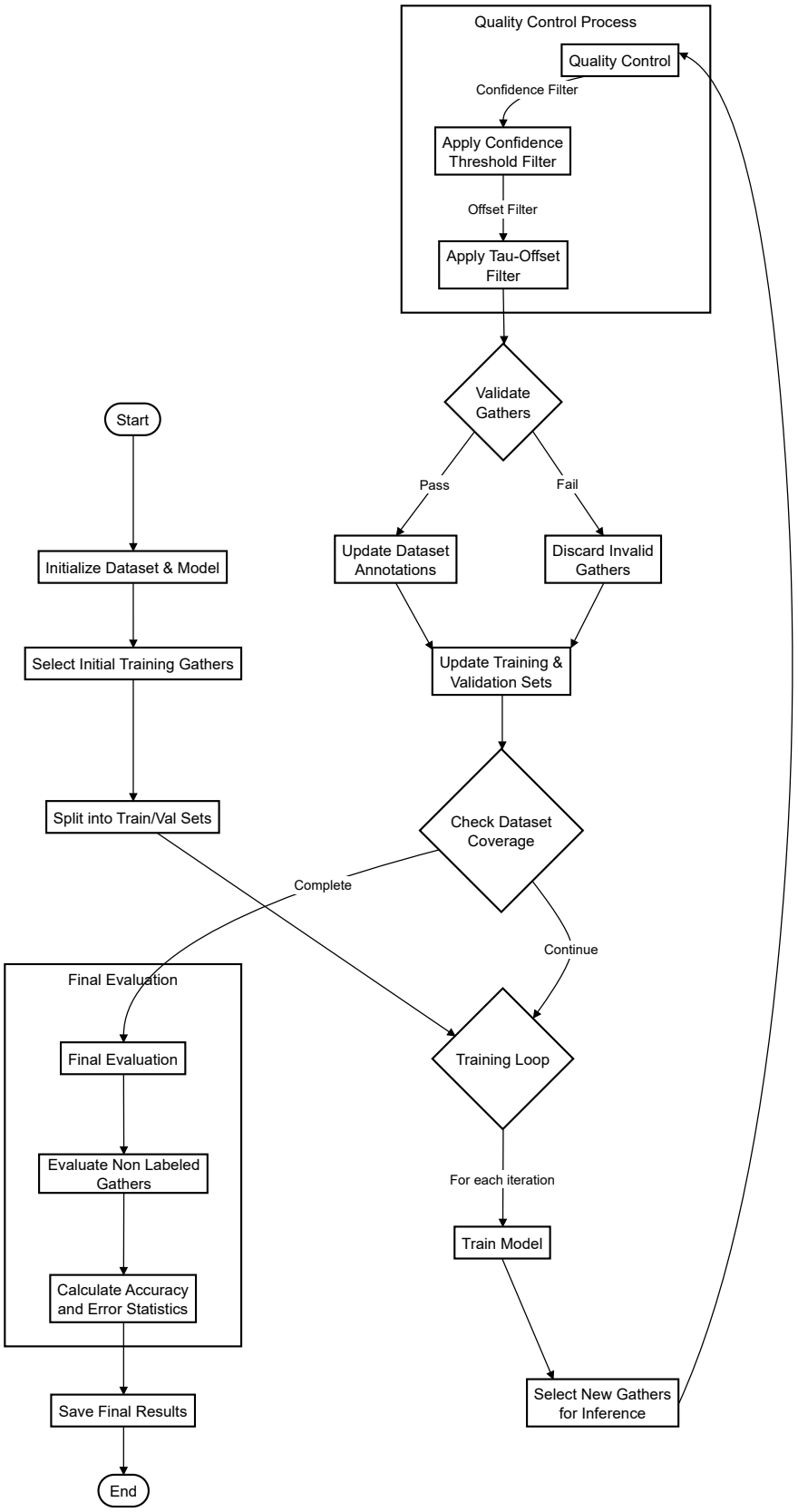

Figure 2 – Iterative Training Workflow, source: author

3.2 Model Architecture

The implementation uses a U-Net architecture for seismic data segmentation. The network processes seismic gathers as input, with dimensions of traces × time samples, applying padding to accommodate varying gather sizes. The architecture follows an encoder-decoder structure with skip connections.

While various deep learning architectures could potentially be employed for this task, we focus on the U-Net architecture to demonstrate our iterative training framework. This choice is motivated by U-Net's proven effectiveness in seismic interpretation tasks, as demonstrated by previous studies (St-Charles et al., 2023; Zwartjes and Yoo, 2022).

Each convolution block in the network consists of two sequential 3×3 convolutions. The convolutions use zero padding with size 1 and stride 1, maintaining spatial dimensions while expanding the feature space. These convolutions operate without bias terms, relying on the subsequent batch normalization layers to center the feature distributions.

Following each convolution, a batch normalization layer standardizes the feature maps, facilitating stable gradient flow during training. The normalized outputs pass through a leaky ReLU activation function with a negative slope of 0.01, allowing for small negative gradients that help prevent dead neurons.

The contracting path applies these double convolution blocks sequentially, with each block followed by a 2×2 max pooling operation. This progression reduces spatial dimensions while increasing feature depth through [64, 128, 256, 512] channels. The expanding path mirrors this structure, using transposed convolutions for upsampling. Skip connections transfer feature maps from the contracting path to the upsampled outputs, maintaining spatial precision through the decoding process. The final layer applies a 1×1 convolution to produce two feature maps matching the input dimensions, representing the background and first-break classes.

3.3 Datasets

The experiments use two mining seismic datasets from Canada: Halfmile, and Sudbury, as provided by St-Charles et al. (2023). The datasets represent land-based reflection seismology surveys, where explosive charges generate shockwaves that propagate through the subsurface. The waves are recorded by geophones arranged in parallel lines across the survey area.

Each dataset consists of line gathers, where a gather represents the collection of traces from a single explosive source activation. When visualized, these gathers form 2D images with the horizontal axis representing the spatial distribution of receivers and the vertical axis representing time samples (Figure 4). The datasets present varying

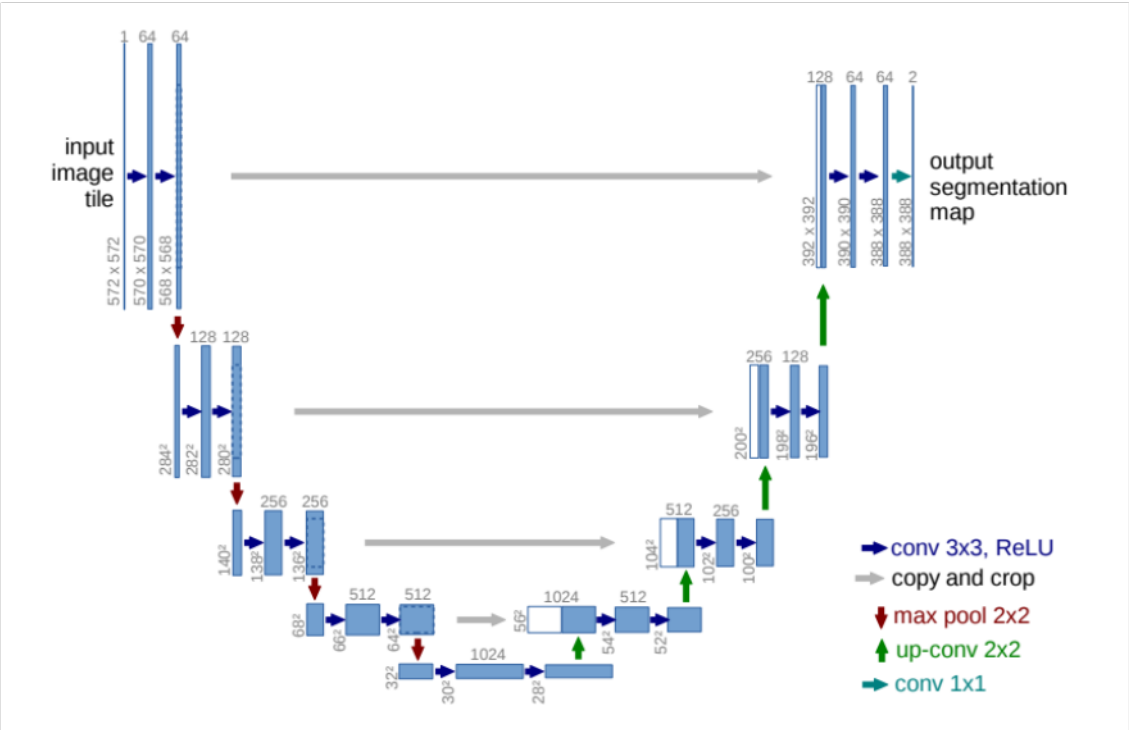

Figure 3 – Standard unet, source: Ronneberger et al. (2015)

characteristics in terms of acquisition parameters:

Table 1 – Characteristics of the seismic datasets

Parameter	Halfmile	Sudbury
Sampling Rate (ms)	2	2
Samples per Trace	751	1001
Number of Shots	690	892
Recording Time (s)	1.502	2.002

The first-break annotations were also provided by St-Charles et al. (2023). These annotations cover approximately 72.3% of traces in the datasets, selected based on signal strength and quality criteria determined by expert analysis. The picks were initially generated using automated tools based on STA/LTA ratios and subsequently underwent manual correction by domain experts.

3.4 Data Preprocessing

The process begins with gather standardization to create uniform input dimensions. Each gather is adjusted to match the dimensions of the largest gather in the dataset

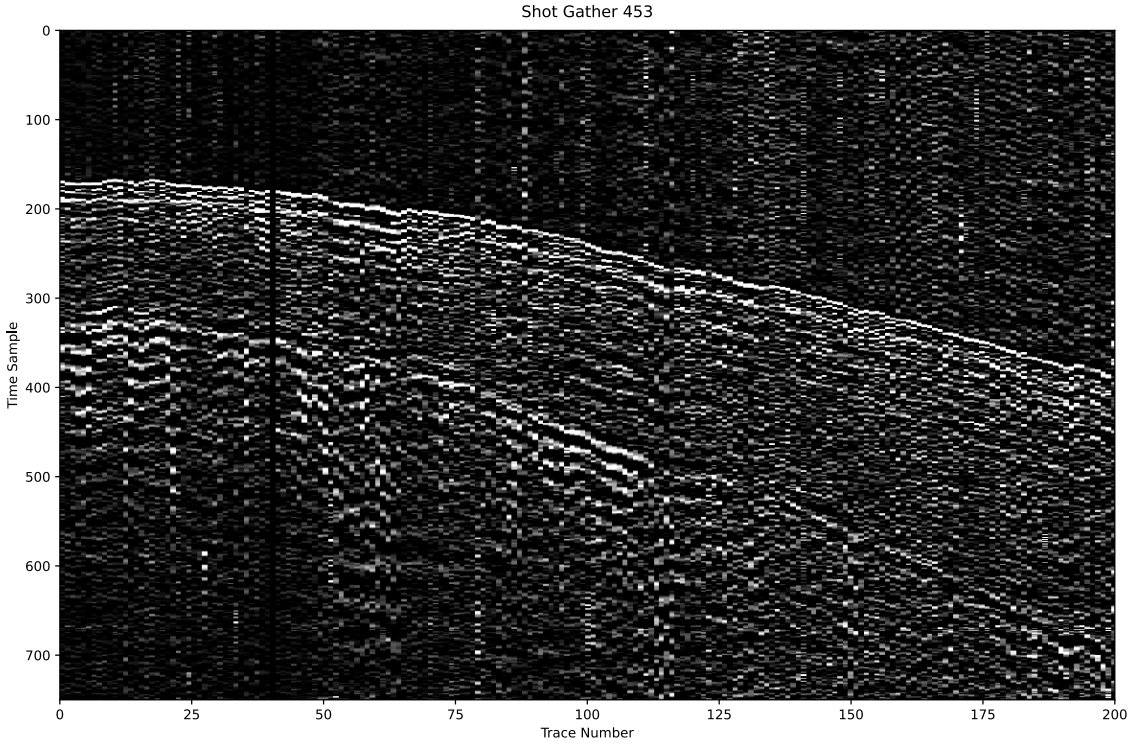

Figure 4 – Line Gather 438, Halfmile dataset. Source: author

using padding values of one. Internal testing demonstrated that this padding strategy outperforms conventional zero-padding and edge-value extension methods in preserving gather characteristics at boundaries.

Previous studies have implemented various data augmentation methods, including random cropping, amplitude nullification, dead trace insertion, and polarity inversion (St-Charles et al., 2023; Mardan et al., 2024; Zwartjes and Yoo, 2022). While these techniques can improve model robustness, this work adopts a more focused preprocessing approach to establish clear baselines and ensure reproducibility.

Amplitude normalization forms a key component of the preprocessing pipeline. The process scales samples to values between zero and 65,535, mapping to the unsigned 16-bit integer range. This quantization serves two purposes: it reduces noise in the signal representation and limits amplitude variation precision. This approach draws from computer vision applications, where three 8-bit channels provide sufficient precision for complex feature extraction tasks.

To address the effects of geometric spreading in seismic data, where wave amplitude decreases with the square of traveled distance, normalization occurs on a per-trace basis. This method effectively implements a gain control, enhancing later arrivals and far-offset signals while maintaining relative amplitude relationships within individual traces.

The preprocessing choices described above were informed by preliminary testing,

though systematic ablation studies could provide additional insights into their relative importance. The focus here was on establishing a reliable baseline preprocessing pipeline that enables effective model training while maintaining computational efficiency.

3.5 Iterative Training Workflow

The workflow implements a self-training approach to expand the training dataset while maintaining a quality control. Beginning with a small set of manually labeled data, the process iteratively incorporates new predictions that meet specific confidence and coherency criteria.

3.5.1 Training Process

The training uses the Adam optimizer with a learning rate of 0.001 and batch size of 4, beginning with 5% of available gathers. Each iteration processes 200 new gathers for potential inclusion in the training set, maintaining a 75% training and 25% validation split. To prevent overfitting while preserving learned features, the weights reset every 5 iterations, with each training cycle comprising 30 epochs. This balance between weight preservation and reset helps maintain model stability while allowing adaptation to new data.

This iterative approach inherently provides some protection against error propagation. Samples included in earlier iterations participate in more training cycles than those added later, giving the original manually-picked examples greater influence over the final model. This weighted influence helps maintain model stability despite the potential accumulation of prediction errors.

3.5.2 Quality Control

The quality control system implements two sequential filtering steps to ensure reliable prediction incorporation:

1. Confidence Filtering: The model outputs undergo softmax transformation to obtain prediction probabilities. The process retains only picks with confidence values exceeding 0.70, ensuring high prediction certainty.

2. Spatial Coherence Analysis: Following the method described in Loginov et al. (2022), the process evaluates predictions through tau-offset crossplots. First-arrival predictions are binned into 20 offset windows per gather, and picks deviating by more than 2 standard deviations from the bin mean are excluded. This step ensures spatial consistency across traces. Figure 5 displays an example of such a plot.

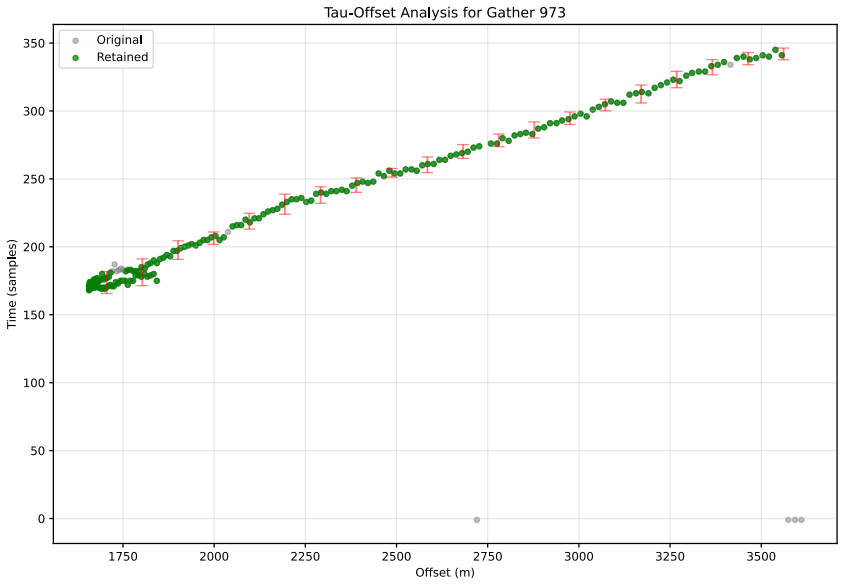

Figure 5 – Tau-offset cross plot for gather 973 (Halfmile). Source: author

The iterative process continues until reaching 15 iterations, ensuring completion of a full 5-iteration training cycle after the final reset. Model evaluation uses only gathers excluded from the initial training set, providing an assessment of the self-training procedure's effectiveness in expanding the labeled dataset.

The threshold values chosen for both filtering steps (0.70 for confidence and 2σ for spatial coherence) were selected based on initial experimentation. While these parameters could be optimized through systematic testing, they provided a reasonable balance between prediction reliability and data retention in our experiments.

The quality control process inherently introduces a selection bias towards picks that conform to expected patterns. While this improves overall reliability, it may exclude valid but unusual picks, particularly in geologically complex areas. For example, in areas with strong lateral velocity variations, valid picks might be rejected by the spatial coherence criteria despite being correct.

3.6 Experimental Setup

For each dataset, we created a region of first breaks by adding a window of 5 samples around the manual picks, transforming the detection problem from a single-sample line to a small region. This modification provides the network with a more stable target during training, though it represents a fundamental trade-off in the picking approach. Traditional first-break picking methods focus on detecting sharp amplitude onsets, as these often represent the true first arrival of seismic energy. By expanding the first-break annotation to a window, we potentially reduce sensitivity to these sharp transitions in

favor of a more general pattern recognition approach.

With a typical sampling rate of 4ms, a 10-sample window represents approximately 0.4 seconds around the first break. This window size was chosen as an acceptable range of uncertainty, considering that variations of this magnitude would not significantly impact subsequent processing steps. The trade-off between onset detection precision and overall robustness is justified by both the improved training stability and this practical tolerance in downstream processing. The iterative training protocol was executed 10 times for each dataset to assess consistency. Table 2 presents the number of gathers used in each stage of the process.

Table 2 – Dataset statistics and training parameters

Dataset	Total Gathers	Initial Gathers	Final Gathers	Iterations
Sudbury	11420	547	5710	15
Halfmile	5520	275	2760	15

To evaluate the methodology, we conducted three sets of experiments:

1. Standard Training with Region: Control experiments using a fixed data split (3.75% training, 1.25% validation, 95% testing) with the first-break region approach. Each run trained for 50 epochs.

2. Standard Training with Single Line: Control experiments using the same fixed split but detecting first breaks as single time samples rather than regions. This configuration helps assess the impact of the region-based approach on model performance.

3. Iterative Training with Region: The proposed iterative methodology using region-based detection, starting with the same initial data split but progressively incorporating new samples through the quality control process.

All experiments maintained identical network parameters, including learning rate, optimizer settings, and model architecture. The only variation between the standard and iterative approaches was the training protocol itself. Each experimental configuration was repeated 10 times. While a larger number of repetitions would be desirable for robust statistical testing, computational and time constraints limited our study to 10 runs per configuration. This number of repetitions still provides useful insights into the variability and general trends of each approach.

For both training approaches, evaluation was conducted on 95% of the gathers not used in the initial training set, enabling direct comparison between methods. While the iterative approach gradually incorporates some of these gathers into its training process, we intentionally evaluate against the original manual picks rather than the model-generated

labels used during training. This evaluation strategy serves two purposes: First, it maintains consistency with the control experiments by assessing both approaches against the same set of expert annotations. Second, it provides a fair assessment of each method's ability to replicate human picking behavior. Using the same evaluation set, despite its partial overlap with the iterative training process, offers a practical measure of how well each approach generalizes from the initial 5% of labeled data to the broader dataset. This overlap is an inherent feature of the iterative methodology rather than a limitation of the evaluation process, as it reflects the method's core strategy of progressively expanding its training set while maintaining pick quality.

All experiments were conducted using an NVIDIA Tesla V100 GPU on the *Núcleo de processamento de Alto Desempenho* (NPAD) at UFRN.

3.7 Evaluation Metrics

The model performance was evaluated using three complementary metrics that capture different aspects of first-break picking accuracy.

The exact pick accuracy (A_{exact}) measures the proportion of predictions that exactly match the original manual picks:

$$A_{\text{exact}} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(y_i = \hat{y}_i) \tag{3.1}$$

where N is the total number of traces, y_i is the manual pick for trace i, $\hat{y}_i$ is the predicted pick, and $\mathbb{1}$ is the indicator function.

The window-based accuracy (A_{window}) considers predictions within a 10-sample window of the manual pick as correct:

$$A_{\text{window}} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(|y_i - \hat{y}_i| \leq 5) \tag{3.2}$$

The mean absolute error (MAE) in samples quantifies the average deviation between predictions and manual picks:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i| \tag{3.3}$$

These metrics were chosen because they provide complementary information about the model's performance. A_{exact} evaluates strict accuracy, useful for assessing precise picking capability. A_{window} accounts for acceptable variation in first-break interpretation,

recognizing that small differences in pick location often do not significantly impact subsequent processing steps. The MAE provides a continuous measure of prediction deviation, offering insight into the magnitude of picking errors when they occur.

Invalid picks (where either the manual pick or prediction is missing or marked as invalid) were excluded from all metric calculations. For evaluation, metrics were calculated on all gathers except those used in the initial training set. While some of these gathers were used during the iterative training process, the evaluation compares predictions against the original manual picks, not the model-generated labels used during training. This evaluation strategy enables direct comparison with the control experiments, as both approaches are assessed against the same set of manual picks, providing a fair assessment of each method's ability to replicate expert annotations.

4 RESULTS AND DISCUSSIONS

This chapter analyzes the experimental results for first-break picking using deep learning models. The evaluation focuses on three key aspects: the effectiveness of window-based first-break detection compared to single-sample approaches, the performance of the iterative training methodology, and detailed error analysis of model predictions. Results are compared across two seismic datasets, Halfmile and Sudbury, to assess the generalizability and robustness of the proposed methods.The analysis encompasses both computational performance characteristics and qualitative assessment of prediction quality through representative gather examples. Special attention is given to the convergence behavior during iterative training and the distribution of prediction errors, offering insights into the models' learning patterns and limitations.

4.1 First-Break Window Region

The introduction of a window region around first breaks aims to account for uncertainty inherent in manual picking and provide a more stable training target. Tables 4 and 3 present the results comparing models trained with and without a first-break window region. The experiments used a fixed data split of 3.75% for training, 1.25% for validation, and 95% for testing across the datasets.

The window-based approach shows lower exact accuracy (A_{exact}) compared to the single-line approach, as the model must identify a specific sample within the target window. The window-based approach achieves lower mean absolute error (MAE) across both datasets: Halfmile (16.16 vs 29.79 samples) and Sudbury (22.69 vs 25.05 samples).

The window-based method demonstrates more stable performance, with lower standard deviations in both MAE and window accuracy (A_{window}). For the Sudbury dataset, the standard deviation of A_{window} decreases from 0.1262 to 0.0367. The standard deviations of the metrics provide insight into method reliability. The window-based approach shows consistent standard deviations across datasets for A_{exact} (0.0172-0.0255), while the single-line approach shows higher variation, particularly for Sudbury (0.1486).

The impact of the window-based approach varies across datasets. Halfmile shows the highest window accuracy (0.9375) while maintaining the lowest MAE (16.16 samples), suggesting that its first breaks may be more consistently defined. In contrast, Sudbury exhibits lower window accuracy (0.8136) despite having moderate MAE values, indicating more complex picking patterns.

The results show a trade-off between A_{exact} and A_{window} metrics. The single-line

approach achieves higher exact matches, while the window-based method produces picks that are more consistently within the acceptable range of true first breaks. This aligns with seismic processing requirements, where small variations in pick location have limited impact on subsequent processing steps. The lower MAE values in the window-based approach suggest better overall pick positioning, even when exact matches are not achieved.

Table 3 – Performance metrics for models trained with single-sample first breaks using 5% of total gathers

	A_{exact}		MAE		A_{window}	
Dataset	Mean	Std	Mean	Std	Mean	Std
Halfmile	0.7498	0.0149	29.7867	4.9438	0.8607	0.0221
Sudbury	0.3065	0.1486	25.0489	11.0476	0.7720	0.1262

Table 4 – Performance metrics for models trained with first-break window using 5% of total gathers

	A_{exact}		MAE		A_{window}	
Dataset	Mean	Std	Mean	Std	Mean	Std
Halfmile	0.1800	0.0172	16.1636	4.4181	0.9375	0.0183
Sudbury	0.1863	0.0255	22.6939	3.6451	0.8136	0.0367

This trade-off reflects a fundamental shift from precise onset detection to pattern recognition. While single-sample training encourages the model to identify sharp amplitude transitions traditionally associated with first breaks, the window-based approach prioritizes the recognition of broader arrival patterns. The lower MAE values across all datasets suggest that this compromise in exact onset detection leads to more robust and reliable picking behavior overall.

Both approaches present similar computational requirements. However, the window-based method requires additional preprocessing to create the target regions. This overhead is minimal compared to the total training time and is performed only once during dataset preparation. Based on these results, the window-based approach was selected for the iterative training experiments, as it provides more stable predictions while maintaining acceptable pick accuracy within practical tolerances.

4.2 Iterative Training

4.2.1 Computational Performance

Training was performed with a batch size of 4, balancing memory constraints with computational efficiency. The training process consisted of 15 iterations with 30 epochs

per iteration, totaling 450 epochs for each experimental run.

Table 5 – Training time statistics

Dataset	Training Time (minutes)	Wall Time (minutes)	Mean Epoch Time (minutes)	Time Std Dev (minutes)
Halfmile	581.42	629.50	1.38	6.47
Sudbury	547.83	590.87	1.30	24.47

For the Halfmile dataset, the average training time per run was 581.42 minutes (approximately 9.7 hours), with a standard deviation of 6.47 minutes. The mean epoch time was 1.38 minutes, demonstrating consistent performance across iterations. The total wall time, including data loading and quality control steps, averaged 629.50 minutes (approximately 10.5 hours).

The Sudbury dataset showed slightly faster processing times, with an average training time of 547.83 minutes (approximately 9.1 hours) and a standard deviation of 24.47 minutes. The mean epoch time was 1.30 minutes. The total wall time averaged 590.87 minutes (approximately 9.8 hours). The difference between training time and wall time, approximately 48 minutes for both datasets, represents the overhead from data loading, quality control processes, and model evaluation steps between iterations.

4.2.2 Convergence

Figures 6, 7, and 8 illustrate the evolution of training and validation loss and accuracy over 15 iterations for a single Sudbury dataset run. Each plot corresponds to the training of a U-Net model over 150 epochs, with iterations grouped into batches of 30 epochs. The number of training examples increases progressively with each iteration as new labeled gathers are incorporated into the training set.

The plots show that training loss consistently decreases across iterations, reflecting the model's capacity to learn from the training data. In contrast, validation loss exhibits occasional increases, particularly in the later iterations (e.g., iterations 11–15 shown in Figure 8), suggesting a pattern of overfitting as the model begins to specialize on the training data. However, validation accuracy remains relatively stable during these iterations and increases over time, indicating that the iterative inclusion of new data supports improvements in performance.

Final model accuracy is computed using all gathers not included in the initial training set. This includes some gathers that were incrementally added to the training set during the iterative process. As a result of this evaluation methodology, the distinction between training, validation, and test sets becomes less rigid, inherently leading to some overlap between the data used for training and final evaluation. While this overlap could be interpreted as a potential source of overfitting, it is a natural consequence of the

chosen evaluation strategy and does not significantly undermine the validity of the results. This setup ensures that the evaluation reflects the model's performance across a broader representation of the available data, including training and validation gathers, thereby reducing the practical impact of overfitting concerns.

Lastly, internal testing indicates that reducing the number of epochs per iteration results in lower performance. While training for a greater number of epochs introduces the risk of overfitting, it also allows the model to develop a deeper representation of the data. The Halfmile dataset displays a similar behaviour.

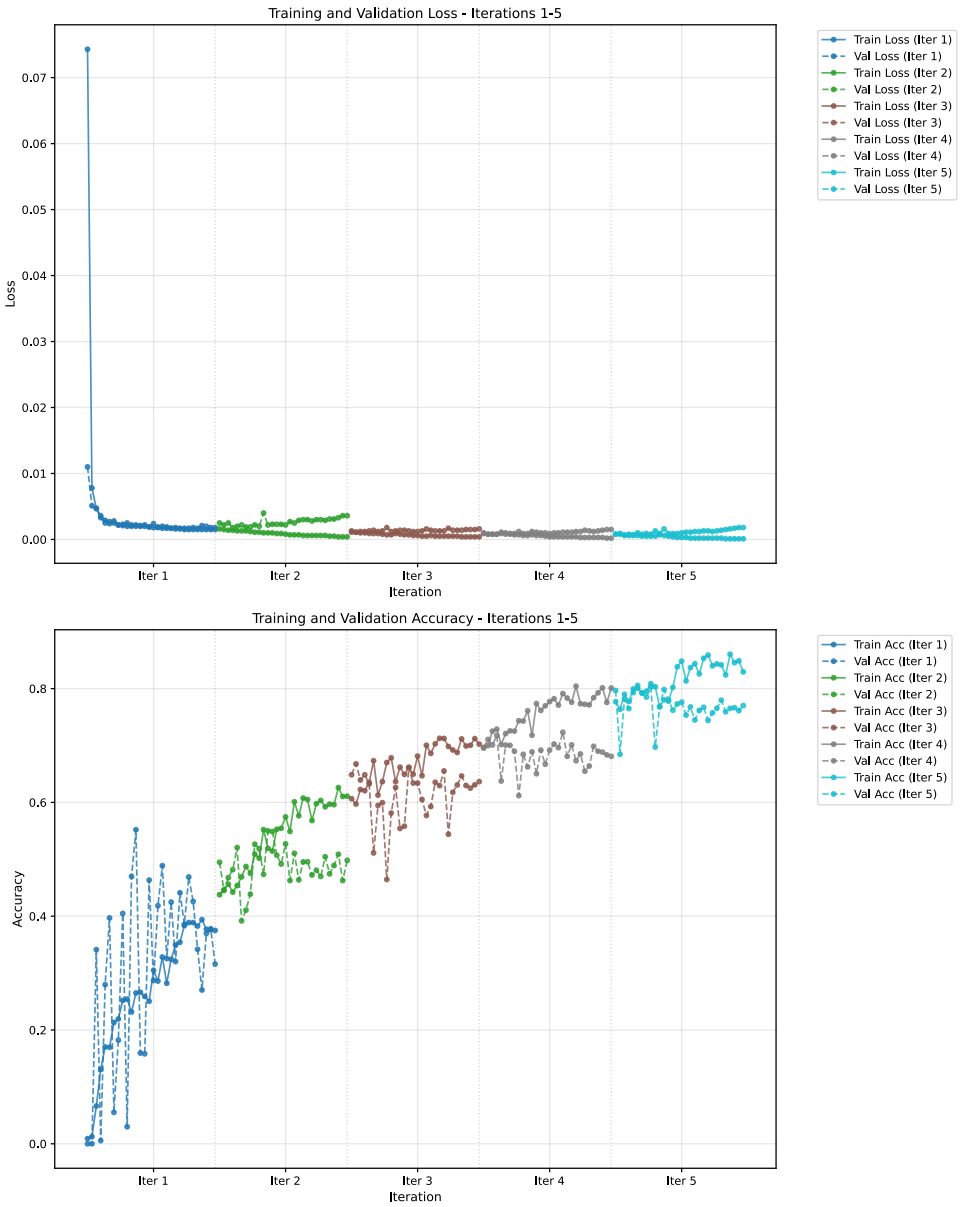

Figure 6 – Training and validation loss (top) and accuracy (bottom) for Sudbury - Iterations 1 to 5. Source: author.

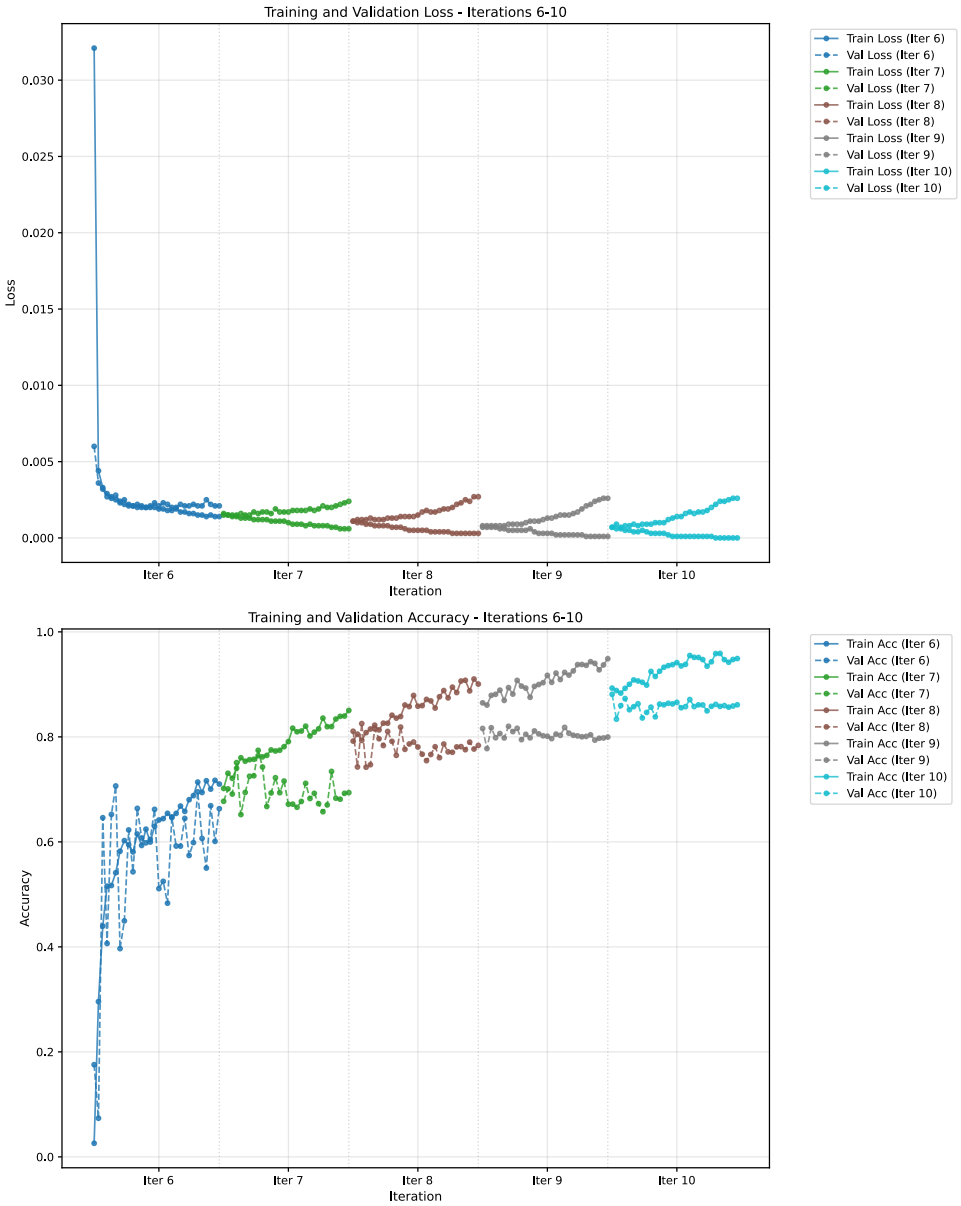

Figure 7 – Training and validation loss (top) and accuracy (bottom) for Sudbury - Iterations 6 to 10. Source: author.

4.2.3 Performance

The iterative training approach demonstrates varying effectiveness across the datasets. For the Halfmile dataset, the iterative method achieves substantial improvements over the standard window-based training. The MAE decreases significantly from 16.16 to 9.16 samples, while the window accuracy (A_{window}) improves from 0.9375 to 0.9677. The lower standard deviations across all metrics (MAE: 1.4104 vs 4.4181, A_{window}: 0.0054 vs 0.0183) suggest that the iterative approach also leads to more consistent model performance.

In contrast, the Sudbury dataset shows a slight degradation in performance with the iterative approach. While the exact accuracy (A_{exact}) remains relatively stable (0.1863 to

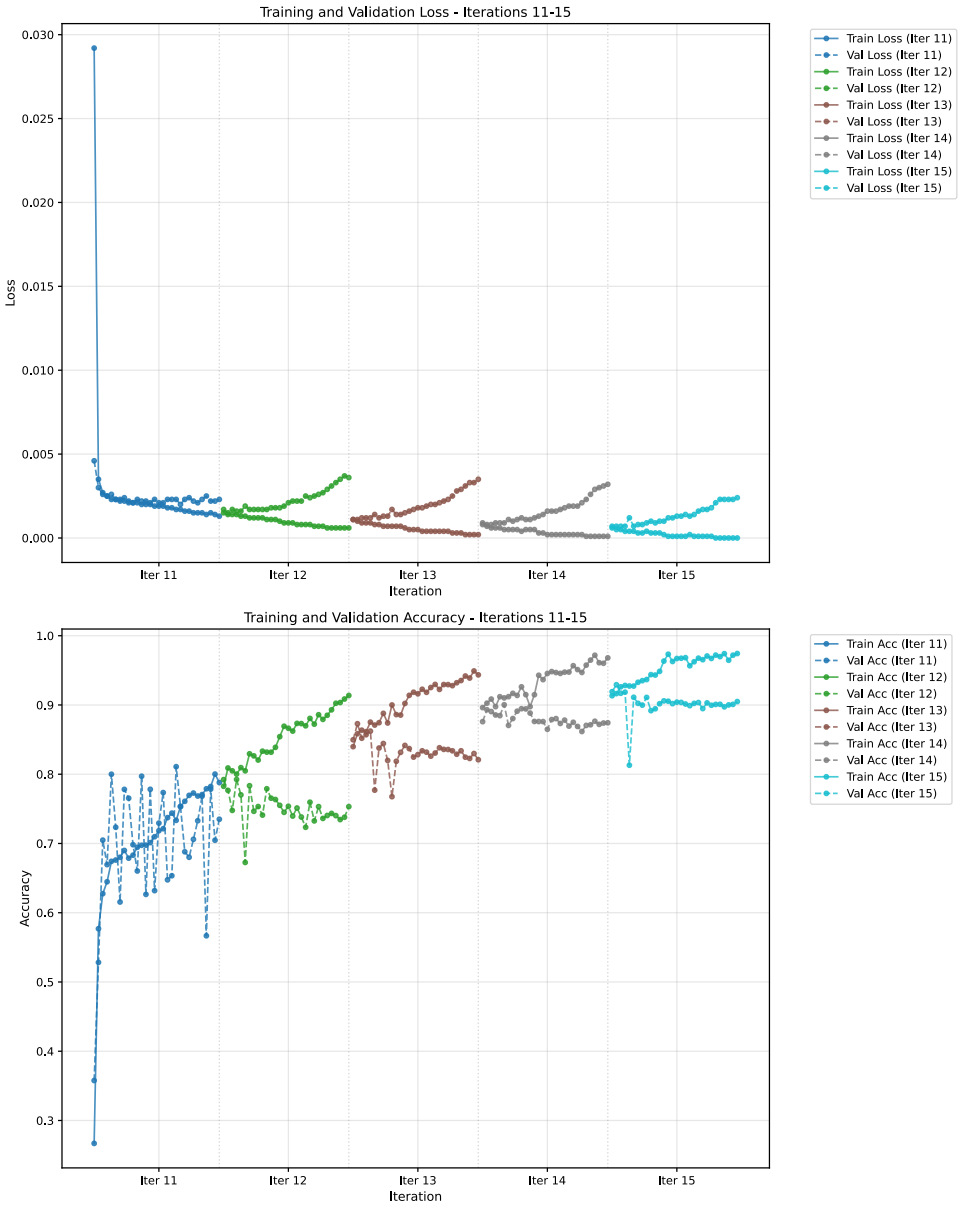

Figure 8 – Training and validation loss (top) and accuracy (bottom) for Sudbury - Iterations 11 to 15. Source: author.

0.1349), the MAE increases from 22.69 to 26.42 samples. However, the standard deviation of the MAE improves from 3.6451 to 2.3790, indicating more consistent predictions despite the increased average error. The window accuracy shows a marginal decrease from 0.8136 to 0.7881, though with improved consistency as indicated by the lower standard deviation (0.0367 to 0.0215).

These results suggest that the effectiveness of the iterative training approach may depend on the inherent characteristics of the dataset. The Halfmile dataset, which showed better initial performance, benefits significantly from the iterative process. This might indicate that datasets with more consistent first-break patterns are better suited for this approach, as the quality control process can more reliably identify valid predictions for

Table 6 – Performance metrics for models trained with first-break window using 5% of total gathers - Iterative Method

	A_{exact}		MAE		A_{window}	
Dataset	Mean	Std	Mean	Std	Mean	Std
Halfmile	0.1420	0.0162	9.1617	1.4104	0.9677	0.0054
Sudbury	0.1349	0.0140	26.4183	2.3790	0.7881	0.0215

inclusion in the training set.

The performance differences can be better understood by examining both the quality control retention rates and the inherent dataset characteristics. For Halfmile, the confidence and offset filtering retain 99.33% (±0.77%) and 97.68% (±0.24%) of picks respectively, resulting in a total retention rate of 97.03% (±0.70%). For Sudbury, these rates are 93.09% (±9.74%) and 97.76% (±0.34%) for offset filtering, with a total retention of 90.99% (±9.38%).

The more modest performance of the iterative approach on the Sudbury dataset, despite reasonable retention rates, can be explained by the underlying data quality issues identified by St-Charles et al. (2023). Their analysis reveals that Sudbury's annotations are of lower quality compared to other datasets like Halfmile, with only 44.7% of line gathers being considered valid. While our methodology excluded completely invalid line gathers, we did not implement the more stringent filtering suggested by their study for "useful" line gathers. This decision may have led to training on picks of questionable quality, potentially explaining both the higher variability in confidence retention (±9.74%) and the slight degradation in performance metrics during iterative training. In contrast, Halfmile's data quality was noted to be consistently higher, with 99.6% of line gathers considered valid, which aligns with its superior performance in our experiments.

4.3 Model Predictions

4.3.1 Error Distribution Analysis

The error distribution analysis for both datasets provides insights into the model's prediction patterns and limitations. Figures 9 through 12 present these distributions at different scales: a detailed view of errors within ±10 samples and a broader perspective spanning ±200 samples.

Within the ±10 sample window, both datasets exhibit approximately symmetric error distributions centered at zero, indicating no systematic bias in the predictions. However, the broader ±200 sample analysis reveals distinct patterns between datasets. The Halfmile dataset maintains concentrated errors near zero, demonstrating consistent

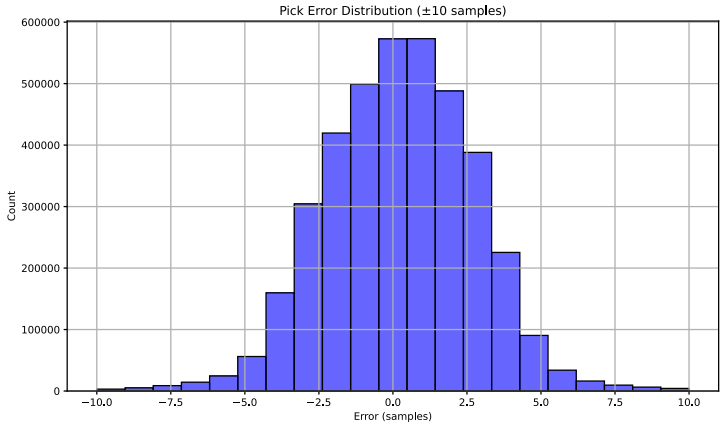

Figure 9 – Picking error distribution - ±10 samples - Halfmile. Source: author

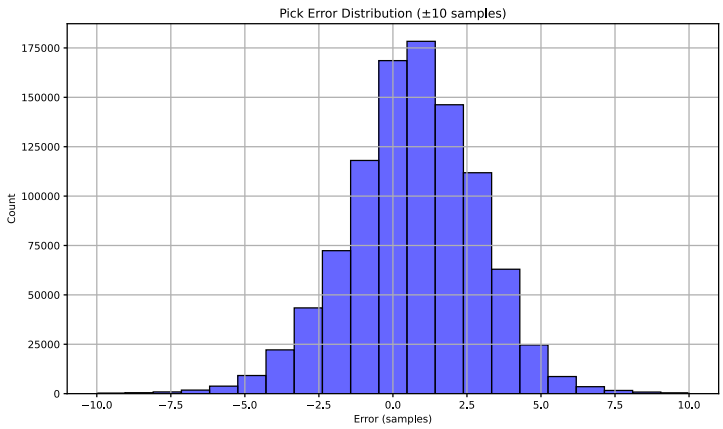

Figure 10 – Picking error distribution - ±10 samples - Sudbury. Source: author

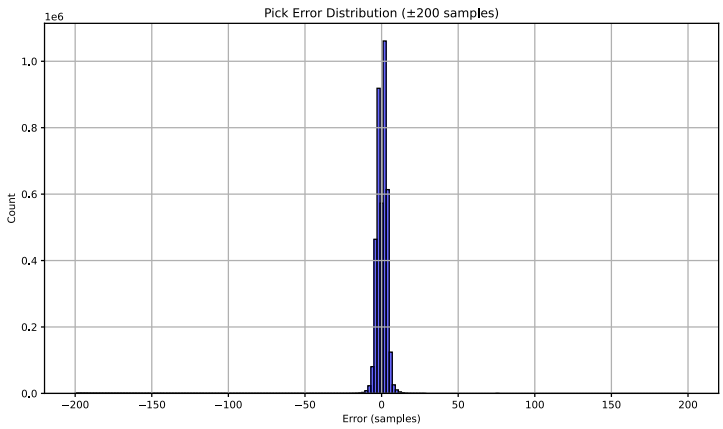

Figure 11 – Picking error distribution - ±200 samples - Halfmile. Source: author

prediction accuracy. In contrast, the Sudbury dataset shows an asymmetric distribution with a tendency toward early predictions, primarily resulting from instances where the model fails to make valid predictions rather than systematic early picking.

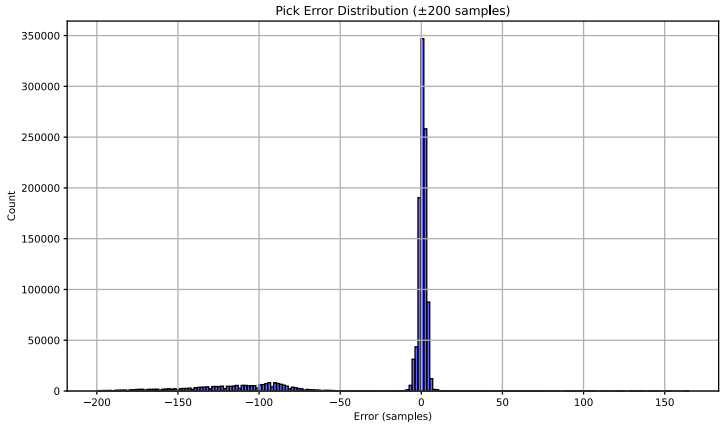

Figure 12 – Picking error distribution - ±200 samples - Sudbury. Source: author

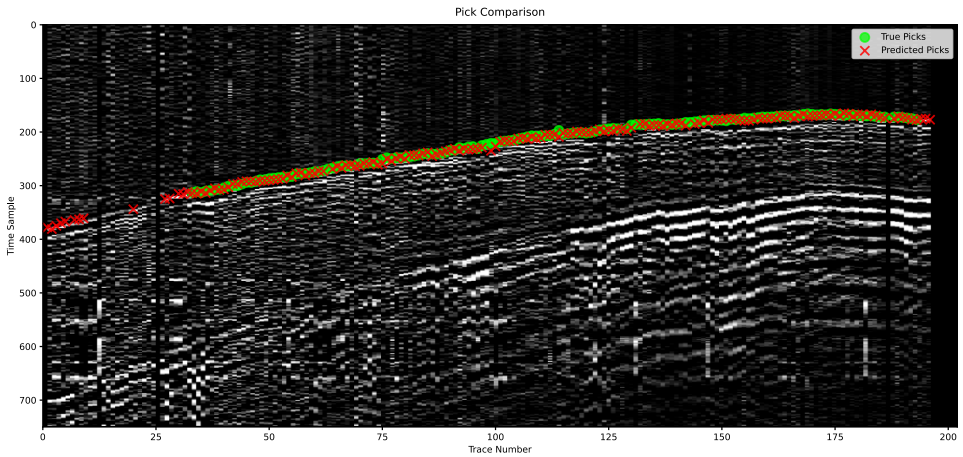

Figure 13 – Model predictions for gather 5088, Halfmile. Source: author

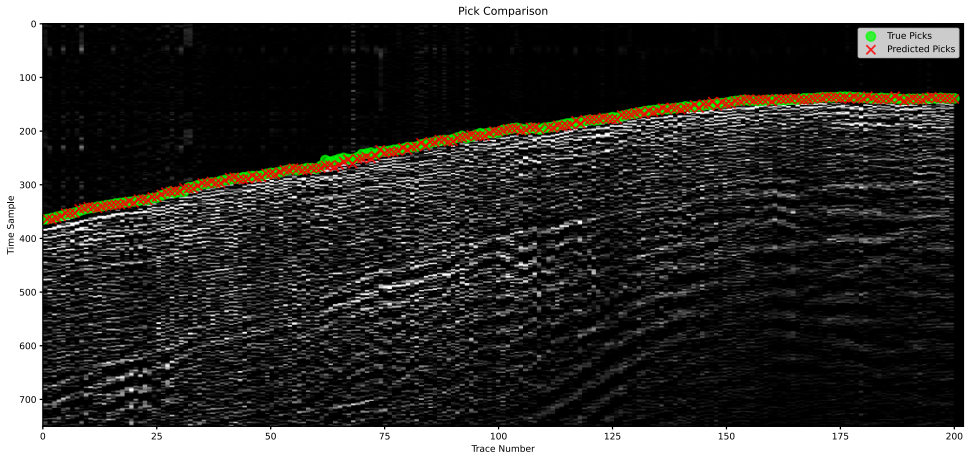

Figure 14 – Model predictions for gather 5146, Halfmile. Source: author

4.3.2 Qualitative Analysis

Figures 13, 14, and 15 demonstrate the model's performance on the Halfmile dataset. The predictions (shown in red) consistently align with the ground truth picks (shown in green), even in the presence of noise. The model maintains picking accuracy across varying

signal quality conditions, suggesting robust feature extraction capabilities for this dataset.

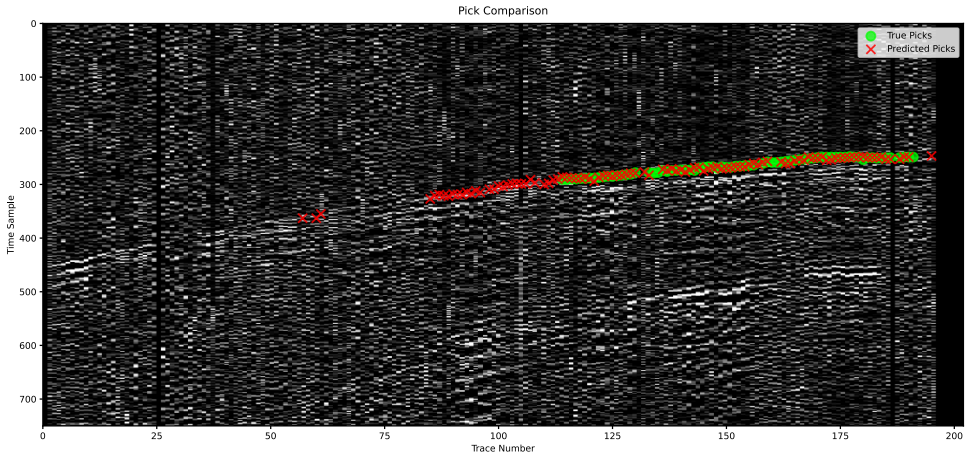

Figure 15 – Model predictions for gather 5280, Halfmile. Source: author

The model's performance on the Sudbury dataset, illustrated in Figures 16 and 17, reveals more complex prediction patterns. Figure 16 demonstrates partial success in pattern recognition, where the model captures the general trend of first breaks but fails to maintain accuracy across all traces. Figure 17 presents an interesting case where the model generates apparently consistent predictions for traces lacking ground truth labels, suggesting potential utility in expanding pick coverage.

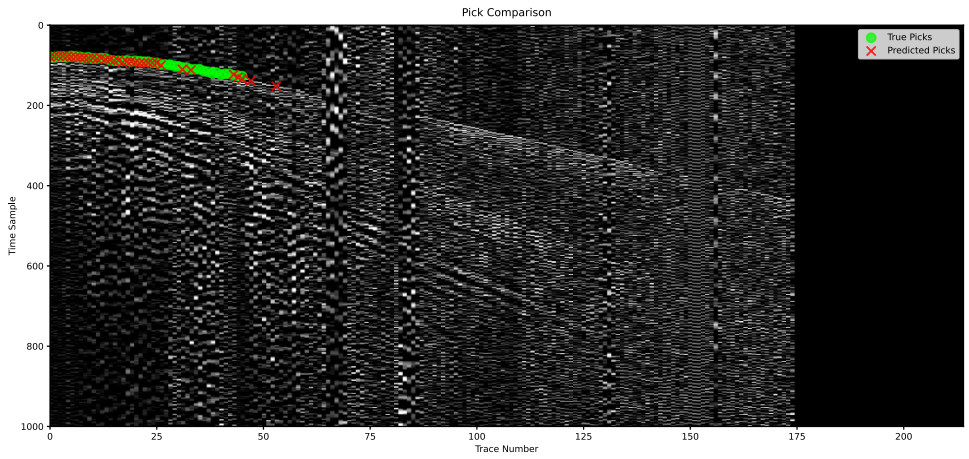

Figure 16 – Model predicions for gather 548, Sudbury. Source: author

This analysis reveals that prediction quality varies between datasets, with Halfmile showing more consistent performance compared to Sudbury. These differences align with the quantitative metrics presented in the previous section and reflect the varying complexity and data quality between the two surveys.

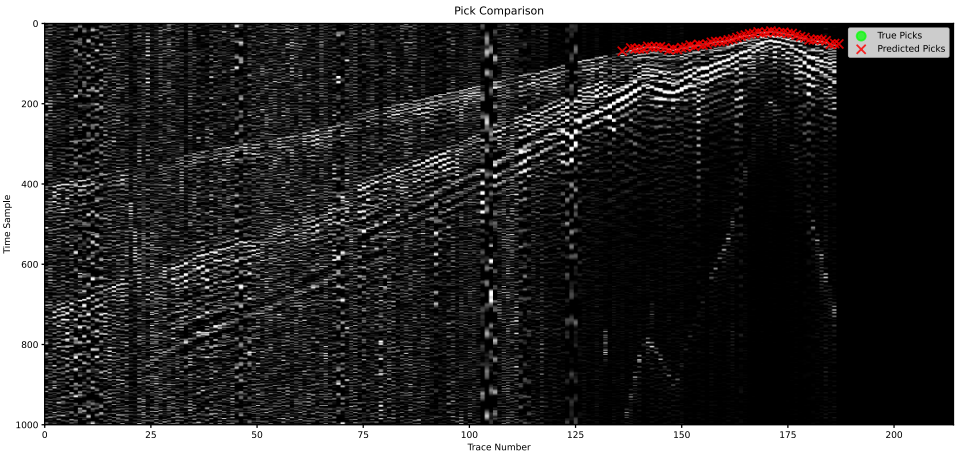

Figure 17 – Model Predcitions for gather 792, Sudbury. Source: author

5 CONCLUSION

This work presented an iterative deep learning framework for first-break detection that prioritizes adaptability to specific seismic datasets over pursuit of universal generalization. The Halfmile dataset showed significant improvements through iterative training, reducing mean absolute error from 16.16 to 9.16 samples while maintaining high window accuracy ($A_{\text{window}} = 0.9677$), while the more modest gains with Sudbury reflect the framework's sensibility to varying data quality.

The contrast in performance between datasets validates our fundamental design philosophy: rather than attempting to create a broadly generalizable model, we focused on developing a workflow that efficiently learns from limited high-quality examples within specific survey contexts. This approach better serves practical requirements in seismic processing, where reliable picks for a particular survey are more valuable than a generalized but potentially less accurate solution.

5.1 Limitations

While these results demonstrate the potential of our adaptable workflow approach, some limitations should be acknowledged that represent opportunities for future research and potential improvements to the methodology.

The number of experimental runs (10 repetitions per configuration) limits the statistical robustness of our results. A larger number of trials would provide more reliable estimates of the model's performance variability and enable more rigorous statistical analysis. This limitation was primarily due to computational resource and time constraints.

Our quality control approach relies on empirically chosen parameters without extensive optimization. The potential biases introduced by the confidence thresholding and spatial coherence filtering were not thoroughly investigated. Alternative QC metrics or approaches might provide better selection criteria for the iterative training process. Furthermore, the interaction between QC parameters and final model performance remains to be systematically analyzed.

Some aspects of the training process warrant further investigation. The impact of weight reset intervals on model convergence and stability was not extensively studied. The chosen training/validation split ratios and learning rate policies were based on common practices rather than systematic optimization. Alternative training schedules might provide better convergence or more stable results.

The framework's sensitivity to preprocessing choices and acquisition parameters was not comprehensively evaluated. While basic trace normalization proved effective, alternative preprocessing approaches might yield better results. The impact of varying acquisition parameters within datasets, such as receiver spacing and sampling rates, on model performance remains unexplored.

5.2 Future Work

Future research should extend this philosophy of adaptable workflows while systematically addressing current limitations. Key areas for investigation include:

1. Comprehensive ablation studies to:
    - Quantify the impact of preprocessing choices
    - Evaluate the necessity of weight reset intervals
    - Assess the sensitivity to training hyperparameters
2. Refinement of quality control mechanisms through:
    - Systematic optimization of confidence and spatial coherence thresholds
    - Development of metrics incorporating geophysical domain knowledge
    - Investigation of alternative validation approaches
3. Improvement of training data quality by:
    - Establishing rigorous criteria for manual pick selection
    - Developing methods to assess ground truth reliability
    - Creating validation protocols for expert annotations
4. Enhancement of the training process via:
    - Optimization of weight reset strategies
    - Investigation of adaptive learning rate schedules
    - Development of dynamic training/validation split criteria

These directions for future research underscore a key insight from our work: the path to better automated first-break picking may not lie in pursuing universal generalization, but rather in developing workflows that can efficiently adapt to specific survey requirements. While our framework demonstrates promising results and clear paths for improvement, its primary contribution may be this shift in perspective—from seeking a single, general

solution to embracing adaptable, context-specific approaches. This philosophy, supported by our experimental results, suggests a practical way forward for reducing manual effort in seismic processing while maintaining high standards for pick quality. As seismic datasets continue to grow in size and complexity, such adaptable workflows will become increasingly valuable for maintaining efficiency without sacrificing reliability.

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